Day 8 Study Notes

Fri 20 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Current electricity (part 1) (75 min) · Maths: Determinants (part 2) (75 min) · Chemistry: Chemical bonding (part 2) (60 min)
Exam	1 h 15 min	The Day 8 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Current electricity (part 1)

Current electricity gives one to two questions per paper. Today covers what happens inside a conductor, resistance and power. Tomorrow covers cells, Kirchhoff's laws and bridges.

1. Current and drift velocity

- $I = \dfrac{dq}{dt} = neAv_d$, where n is the number of free electrons per unit volume, A the cross-section and v_d the drift velocity.
- $v_d = \dfrac{eE\tau}{m}$, where τ is the relaxation time. Drift speeds are tiny, around 10^{-4} m/s.
- Mobility: $\mu = \dfrac{v_d}{E} = \dfrac{e\tau}{m}$, in $\mathrm{m^2\,V^{-1}\,s^{-1}}$.
- Current density: $J = \dfrac{I}{A} = nev_d = \sigma E$.

2. Resistance and resistivity

- Ohm's law: $V = IR$. $R = \dfrac{\rho l}{A}$, with resistivity $\rho = \dfrac{m}{ne^2\tau}$ and conductivity $\sigma = \dfrac{1}{\rho}$.
- Resistivity depends only on the material and its temperature, not on the shape or size of the conductor.
- **Stretching a wire** keeps its volume fixed, so $R \propto l^2$. Doubling the length makes the resistance four times. For a stretch of $x\%$, the new resistance is $\left(1 + \dfrac{x}{100}\right)^2$ times the old.
- **Temperature:** $R_T = R_0(1 + \alpha\,\Delta T)$. For metals α is positive. For semiconductors it is negative. Alloys such as manganin and constantan have very small α, so they are used for standard resistors.
- Conductors that obey Ohm's law give a straight-line V–I graph. Diodes and other semiconductor devices do not.

3. Combinations of resistors

	Series	**Parallel**
Same for all	current	potential difference
Equivalent	$R = R_1 + R_2 + \dots$	$\dfrac{1}{R} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \dots$
Two resistors	$R_1 + R_2$	$\dfrac{R_1 R_2}{R_1 + R_2}$
n equal resistors	nR	$\dfrac{R}{n}$

4. Energy and power

- $P = VI = I^2R = \dfrac{V^2}{R}$. Heat produced in time t: $H = I^2Rt$. One kilowatt hour $= 3.6 \times 10^6$ J.
- A device rated at power P and voltage V has resistance $R = \dfrac{V^2}{P}$. Used at a different voltage V', it draws $P' = P\left(\dfrac{V'}{V}\right)^2$.
- Bulbs in **series** carry the same current, so the one with more resistance (the lower wattage rating) is brighter.

- Bulbs in **parallel** have the same voltage, so the higher wattage bulb is brighter.

Solved example 1. A wire of cross-section $10^{-6}\ \text{m}^2$ carries 1.6 A. Find the drift velocity if $n = 8 \times 10^{28}\ \text{m}^{-3}$.

$$v_d = \frac{I}{neA} = \frac{1.6}{8 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}} = 1.25 \times 10^{-4}\ \text{m/s}.$$

Solved example 2. A wire is stretched to increase its length by 10%. Find the percentage increase in resistance. New resistance $= (1.1)^2 = 1.21$ times the old, an increase of 21%.

Traps. When a wire is stretched, both length and area change; use $R \propto l^2$ instead of changing only l. For small percentage changes the resistance change is about twice the length change, but for 10% or more use the exact square. In series, the bulb with the lower rating glows brighter, which is the opposite of the parallel case.

Read in NCERT: Physics Class 12 Part 1, chapter "Current Electricity": from the start up to electrical energy and power.

Mathematics · Determinants (part 2)

The results $|\text{adj } A| = |A|^{n-1}$ and the conditions for a system to have no solution or infinitely many are asked in almost every session.

1. Adjoint and inverse

- The adjoint, $\text{adj } A$, is the transpose of the matrix of cofactors.
- For a 2×2 matrix, swap the diagonal entries and change the signs of the other two: $\text{adj} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
- $A\,(\text{adj } A) = (\text{adj } A)\,A = |A|\,I$.
- $A^{-1} = \dfrac{\text{adj } A}{|A|}$. It exists only when $|A| \neq 0$; such a matrix is called non-singular.

2. Results for an $n \times n$ matrix

Result	For $n = 3$
$\|\text{adj } A\| = \|A\|^{n-1}$	$\|A\|^2$
$\text{adj}(\text{adj } A) = \|A\|^{n-2} A$	$\|A\|\,A$
$\|\text{adj}(\text{adj } A)\| = \|A\|^{(n-1)^2}$	$\|A\|^4$
$\text{adj}(kA) = k^{n-1}\,\text{adj } A$	$k^2\,\text{adj } A$
$\|A^{-1}\| = \dfrac{1}{\|A\|}$	
$(AB)^{-1} = B^{-1}A^{-1}$ and $(A^T)^{-1} = (A^{-1})^T$	

> **Solved example 1.** A is 3×3 with $|A| = 3$. Find $|\text{adj}(2A)|$.
> $\text{adj}(2A) = 2^2\,\text{adj } A = 4\,\text{adj } A$. So $|\text{adj}(2A)| = 4^3\,|\text{adj } A| = 64 \times 3^2 = 576$.

3. Inverse from a matrix equation

If a matrix satisfies an equation such as $A^2 - 4A + I = O$, multiply through by A^{-1}: $A - 4I + A^{-1} = O$, so $A^{-1} = 4I - A$.

4. Systems of linear equations, $AX = B$

Condition	Conclusion
$\|A\| \neq 0$	unique solution, $X = A^{-1}B$
$\|A\| = 0$ and $(\text{adj } A)B \neq O$	no solution (inconsistent)
$\|A\| = 0$ and $(\text{adj } A)B = O$	infinitely many solutions, or none

In terms of determinants (Cramer's rule), with $\Delta = |A|$ and $\Delta_x, \Delta_y, \Delta_z$ obtained by replacing a column with the constants: a unique solution when $\Delta \neq 0$; no solution when $\Delta = 0$ and at least one of $\Delta_x, \Delta_y, \Delta_z$ is non-zero.

A **homogeneous** system $AX = O$ always has the zero solution. It has non-zero solutions exactly when $|A| = 0$.

Solved example 2. For what values of λ and μ does the system $x+y+z=6$, $x+2y+3z=10$, $x+2y+\lambda z=\mu$ have no solution?
Subtracting the second equation from the third gives $(\lambda-3)z=\mu-10$. If $\lambda=3$ and $\mu\neq 10$ this reads $0=\text{non-zero}$, so there is no solution. If $\lambda=3$ and $\mu=10$ there are infinitely many. If $\lambda\neq 3$ the solution is unique.

Traps. $|\operatorname{adj} A|$ is $|A|^{n-1}$, so for a 3×3 matrix it is the square, not the cube. Taking a scalar out of an adjoint gives k^{n-1}, and taking it out of a determinant gives k^{n}; questions combine the two. For "no solution" you need $\Delta=0$ together with an inconsistent right-hand side.

Read in NCERT: Mathematics Class 12 Part 1, chapter "Determinants": adjoint and inverse of a matrix, and applications to systems of linear equations.

Chemistry · Chemical bonding (part 2)

Molecular orbital theory supplies the numerical questions of this chapter: bond order and the number of unpaired electrons. Hybridisation questions are solved in seconds with the steric number.

1. Valence bond theory

- A sigma (σ) bond forms by head-on overlap along the line joining the nuclei. A pi (π) bond forms by sideways overlap and is weaker.
- A single bond is one σ. A double bond is one σ and one π. A triple bond is one σ and two π.

2. Hybridisation

Steric number = (number of σ bonds on the central atom) + (number of lone pairs on it).

Steric number	Hybridisation	Examples
2	sp	$BeCl_2$, CO_2, C_2H_2
3	sp^2	BF_3, SO_2, SO_3, C_2H_4, CO_3^{2-}, NO_3^-
4	sp^3	CH_4, NH_3, H_2O, NH_4^+
5	sp^3d	PCl_5, SF_4, ClF_3, XeF_2
6	sp^3d^2	SF_6, BrF_5, XeF_4
7	sp^3d^3	IF_7

More s-character ($sp > sp^2 > sp^3$) means a larger bond angle, a shorter bond and a more electronegative carbon.

3. Molecular orbital theory

Energy order of orbitals for molecules with up to 14 electrons (B_2, C_2, N_2):

σ1s, σ*1s, σ2s, σ*2s, ($\pi 2p_x = \pi 2p_y$), $\sigma 2p_z$, ($\pi^* 2p_x = \pi^* 2p_y$), $\sigma^* 2p_z$

For O_2 and F_2, $\sigma 2p_z$ comes before the π2p pair.

$$\text{Bond order} = \frac{N_b - N_a}{2}$$

N_b and N_a are the numbers of electrons in bonding and antibonding orbitals. A higher bond order means a shorter, stronger bond. A species with unpaired electrons is paramagnetic.

Species	Total electrons	Bond order	Unpaired electrons
H_2	2	1	0
He_2	4	0 (does not exist)	0
B_2	10	1	2
C_2	12	2	0
N_2	14	3	0

N_2^+	13	2.5	1
O_2^+	15	2.5	1
O_2	16	2	2
O_2^-	17	1.5	1
O_2^{2-}	18	1	0
F_2	18	1	0

- Species with the same number of electrons have the same bond order: N_2, CO, NO^+ and CN^- all have 14 electrons and bond order 3.
- Bond length order follows the reverse of bond order: $O_2^+ < O_2 < O_2^- < O_2^{2-}$.

Shortcut for bond order from the total electron count. For 14 electrons the bond order is 3. Each electron added or removed from 14 lowers it by 0.5: 13 or 15 electrons give 2.5, 12 or 16 give 2, 11 or 17 give 1.5, 10 or 18 give 1.

4. Hydrogen bonding

- Forms when hydrogen is bonded to F, O or N. Strength: H–F···H > O–H···O > N–H···N.
- **Intermolecular** hydrogen bonding (HF, water, alcohols, p-nitrophenol) raises boiling point and solubility in water. It is why ice is less dense than water.
- **Intramolecular** hydrogen bonding (o-nitrophenol, salicylaldehyde) happens within one molecule, so the molecules do not hold on to each other: lower boiling point and more volatile.

Traps. O_2 is paramagnetic even though all its electrons look paired in a Lewis structure. B_2 is paramagnetic and C_2 is diamagnetic, because the π2p orbitals fill before σ2p_z. Count lone pairs when finding hybridisation: XeF_4 has steric number 6, not 4.

Read in NCERT: Chemistry Class 11 Part 1, chapter "Chemical Bonding and Molecular Structure": valence bond theory, hybridisation, molecular orbital theory and hydrogen bonding.